Tower Semiconductor (TSEM)

Full investment research | Model assumptions and source evidence

Engineering review: October 6, 2026 | Financial model: October 3, 2026 | Framework 8.0 | Complete current investment report

Status: NOT HELD in the supplied original. **Classification:** bottleneck candidate - qualified merchant photonics capacity. **Artifact analytical Base: 75/100 (45/60)**. **Protected portfolio Excel Base: 99/100**, retained as a separate user override.

Active Q4 2027 operating price target: USD 215.635303/share, pricing Q4 2027-Q3 2028 at the retained **20x forward-four-quarter consolidated GAAP operating profit** and **116.827000 million price target-quarter diluted shares**. The engineering review leaves all operating assumptions, share denominators and both Base scores unchanged.

Tower's forward case already incorporates a much larger photonics business than its reported results. This update tests the October 2 expert capacity, competition and yield claims against that existing path. A nearly unchanged active price target reflects quantified positive and negative operating effects after overlap removal; it does not mean the evidence was ignored. The 2028 management model remains the reference calibration. Current probability-weighted totals are shown separately throughout.

Production research edition as of October 6, 2026: G:/My Drive/1amInvesting/Artifacts/TSEM_TowerSemiconductor_DeepDive.pdf. Portfolio prices and return calculations remain in the master workbook.

1. Supercycle, supply chain and durable advantage

Tower is a specialty foundry with a strong qualified photonics franchise. Customer commitments and manufacturing know-how support the case, but merchant access is not a monopoly: GlobalFoundries and other foundries compete. The relevant economics are wafer realization, product mix, utilization and qualified capacity. Optical form factors can replace one another, so the model does not stack every architecture as separate end-demand.

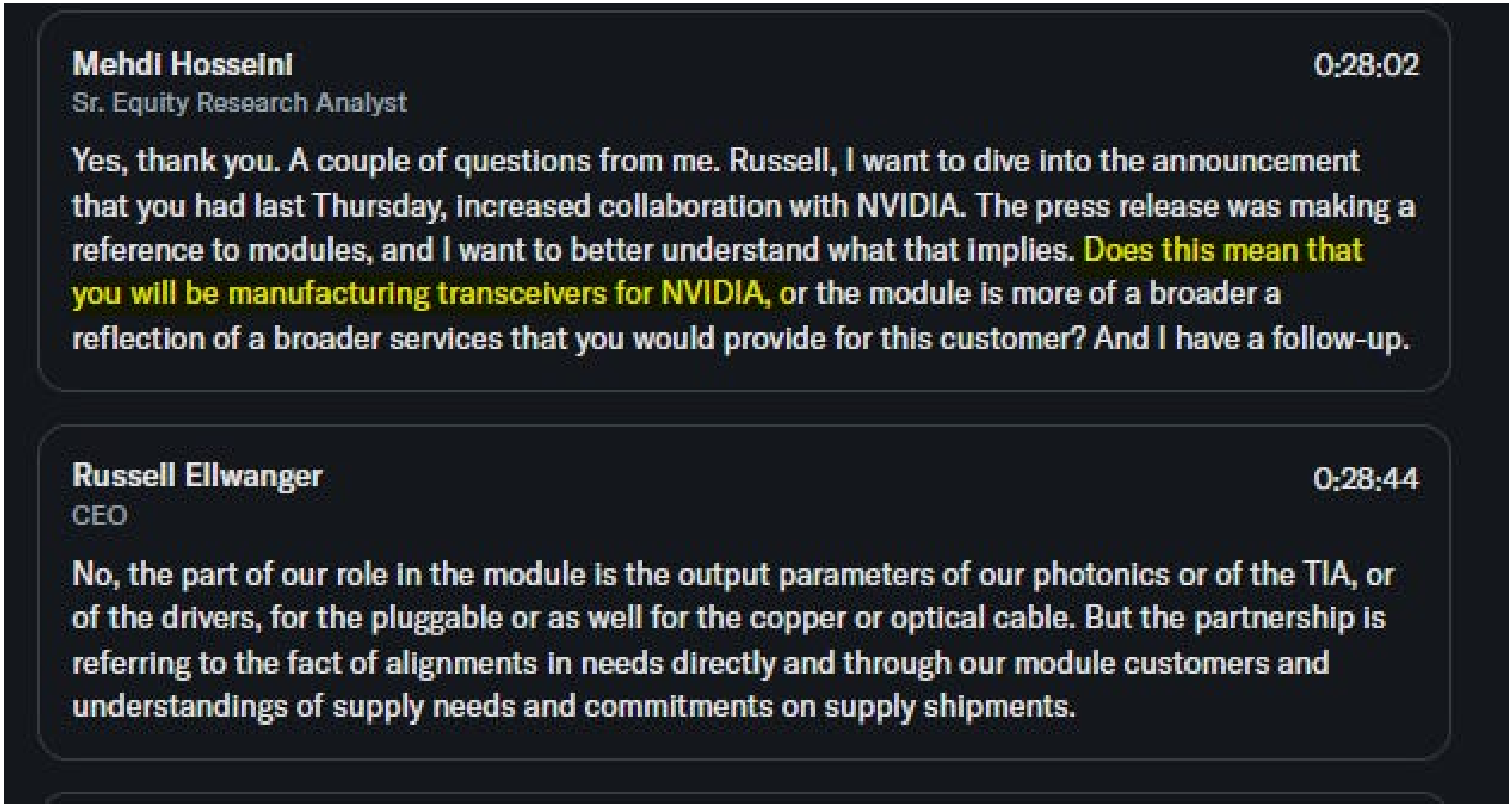
DB1 photonics: growing optical bandwidth → customer PIC designs and optical integration → qualified silicon-photonics/SiGe fabrication → optical engines and systems. Process control, insertion loss, yield and qualification support switching costs. Foundry access is valuable, but merchant availability does not establish sole-source control across the industry. In-house bonding improves production control without creating a separate packaging TAM invoice.

Business / curve	Stage	Evidence and next inflection
SiPho pluggables	Commercial ramp;10-50% qualitative	Current shipments and 2027 commitments support capacity conversion.
NPO and integrated lasers	Early commercial ramp	NewPhotonics volume products provide evidence beyond development plans.
Japan Track 2	Construction/qualification stage	Later installed capacity must become accepted shipments.
RF mobile and legacy analog	Transition/recovery	Process migration can depress current sales before new design wins mature.

Stage percentages are qualitative framework bands relative to a possible mature opportunity, not measured market penetration. A current capacity constraint can coexist with a larger long-term TAM; qualification and economic capture decide how much reaches this company.

Engineering: qualified wafer capacity is the scarce product

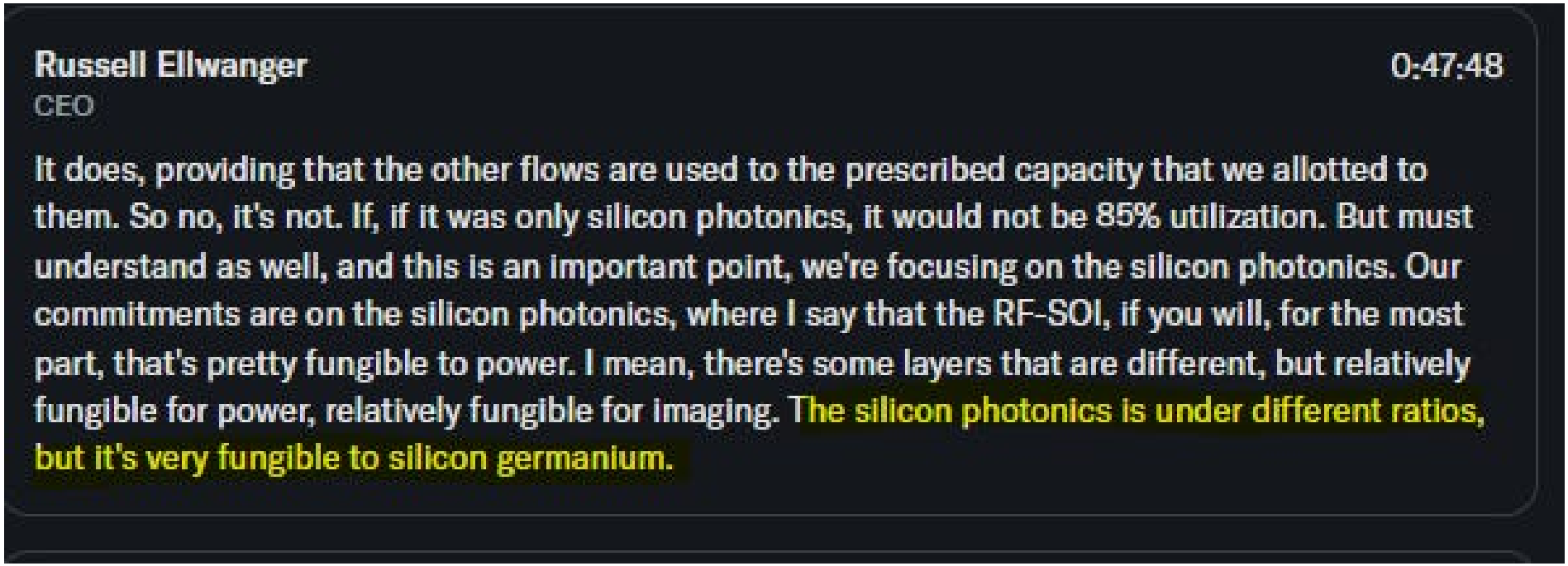
The February 12 Irrational Analysis discussion supports a useful distinction: Tower sells fabricated photonic chips to its customers, including module suppliers; participation in an NVIDIA-compatible optical platform does not mean every PIC is invoiced directly to NVIDIA. The February 5 issuer announcement concerns compatibility with NVIDIA's 1.6T networking protocols. The economic bridge remains accepted wafer/PIC output, realized process pricing and manufacturing cost. Applying a complete transceiver selling price to Tower wafer volume would mix two invoices. [E-TSEM1, E-TSEM2]



Engineering exhibit tower-I19. Saved source: (Market Memo) Tower+GloFo Earnings, HBM4 Qualification Noise, 2026-02-12. Attribution and scope are discussed below.

February management-call excerpt saved in the article: the customer/invoice boundary matters. Tower's supplied photonic chips and its customer's finished optical module are different products.

SiPho and SiGe can share meaningful parts of a manufacturing infrastructure, but shared tools do not make every process step, wafer start and qualified customer flow interchangeable. A line can have spare nameplate capacity while its scarce process module, test step or customer-qualified route is full. More photonic integration can raise process content and die area; poorer yield can consume extra starts without increasing paid output. Conversely, a process improvement can increase good units from the same nominal factory. Those mechanisms belong in realized output and cost, not a second demand multiplier.



Engineering exhibit tower-I22. Saved source: (Market Memo) Tower+GloFo Earnings, HBM4 Qualification Noise, 2026-02-12. Attribution and scope are discussed below.

Historical management excerpt on SiPho/SiGe fungibility. It describes manufacturing flexibility, not evidence that all Tower capacity can instantly become qualified photonics output.

The author's expectation of stronger pricing is directionally consistent with scarce qualified supply and the customer's cost of waiting. It is not a disclosed universal wafer-price increase. Fixed-price commitments, new orders, repricing rights, customer advances and expedite terms must be separated. The February release already committed substantial SiPho/SiGe expansion; the newer May contracts and July/August capacity disclosures supersede that early investment narrative. The current model already subtracts the full SiPho reference-including contracts, Japan and integrated optics-before adding a capacity outcome. [E-TSEM3; T2/T4/T12]

Probability and earnings bridge retained. Capacity acceleration remains **25%**, competitive pricing without that acceleration **20%**, and reference execution **55%**; the separate early yield-cost stress remains **20%**. The capacity outcome applies 90% billing realization to explicitly sized 200mm and 300mm wafer pools, charges 55% marginal COGS plus the retained opex ratio, and subtracts all existing SiPho revenue. The pricing outcome applies its existing bounded concession only to the eligible pool. February's qualitative scarcity argument supplies no incremental signed volume, newly repricable cohort or cost curve beyond these later assumptions. Its additional revenue and OP are therefore zero after overlap; the same-window **Q4 2027 price target remains $215.635303** at 20x forward-four-quarter GAAP OP and 116.827M shares.

The article also speculates about Intel's motivation and possible contract remedies. Tower's February disclosure establishes a dispute and a decision to redirect investment, not a quantified collectible breakup payment. No litigation recovery, foundry revenue or extra net cash is added from that speculation. Nor does criticism of GlobalFoundries establish a Tower share gain: a matched customer, process, qualification result and allocated wafer quantity are needed to determine the amount and

timing of substitution. Credible channel evidence can support a bounded outcome before a formal order, but this historical essay does not supply those missing quantities.

What would move the thesis: evidence of faster accepted output, a new qualified allocation or an actual pricing cohort can increase the active probability-weighted bridge; lost allocations, weaker good-wafer yield, unfavorable contract resets or extra ramp cost reduce it. A generic probe-contact or de-embedding discussion in the same article does not establish a memory manufacturer's HBM qualification outcome and creates no additional Tower revenue.

Engineering sources, reviewed October 6, 2026: E-TSEM1: Irrational Analysis, February 12, full body and available images reviewed; the sole absent image is a small social-profile avatar. E-TSEM2: Tower/NVIDIA announcement, February 5. E-TSEM3: Tower FY2025 release, February 11. Later issuer and expert sources remain dated separately in the source register.

2. Forward TAM and the three-phase expansion lens

Phase	Engineering opportunity	Demand-to-company bridge
Phase 1;100% thematic	800G/1.6T optical PICs	Bandwidth and link growth pull wafer area and process content.The 2027 reference model of 1.4B SiPho includes 1.3B contracted delivery revenue, rather than adding it on top.
Phase 2;50% thematic	NPO, coherent optics and heterogeneous integration	Integrated lasers/modulators broaden invoice content while potentially replacing external components. NewPhotonics' 6.4T roadmap and actual 800G/1.6T shipments sit inside the same SiPho vector.
Phase 3;20% thematic	More advanced optics and Japan Track 2	Planned 20-25k monthly SiPho wafer capacity can support a larger served market after qualification. Later-year capture is explicitly bounded by conversion, not the mere announcement of fourfold Japanese output.

The 100% / 50% / 20% thematic weights guide thesis assessment. They are not blanket discounts to reported sales, contracts or all forecasts. Programme probabilities below govern only separately identified optionality. Current reported sales calibrate the starting point; they do not cap the future market.

3. Current earnings and management evidence

Q2 2026 revenue was USD 460.079M, GAAP GP 137.772M and GAAP OP 90.290M. Diluted weighted-average shares were 114.427M. GAAP diluted EPS was USD 0.79; adjusted EPS was USD 0.88. Q3 revenue guidance is USD 520M +/-5%. The company targets SiPho revenue above a USD 1B annualized rate in Q4 2026.

The August 4 Q2 management presentation and full Q&A were reviewed, with selected May 13 Q1 transcript checks. The September 10 conference was announced, but a complete public transcript was not obtained. September 17 NewPhotonics production news is included. Q2 describes the 2028 model as achievable at minimum on a run-rate basis and nominally for the full year; the retained reference explicitly models the full-year successful-execution case; the October 3 branch-weighted operating schedule is shown below.

Full August 4 prepared remarks and Q&A were reviewed; May 13 selected passages were checked. A complete September 10 conference transcript was not obtained. Product-level costs, final Track 2 economics and post-restructuring minority attribution remain unresolved; none is presented as a disclosed fact.

Q2'26 P&L vs. Q1'26 P&L (in thousands of $)

	Three Months ended June 30, 2026	Three Months ended March 31, 2026
REVENUE	**460,079**	**413,631**
COST OF REVENUE	322,307	302,680
GROSS PROFIT	**137,772**	**110,951**
Research and development	23,569	23,530
Marketing, general and administrative	23,913	22,856
OPERATING PROFIT	**90,290**	**64,565**
Financing and other income, net	15,902	9,518
PROFIT BEFORE INCOME TAX	**106,192**	**74,083**
INCOME TAX EXPENSE, NET	(15,573)	(6,518)
NET PROFIT	**90,619**	**67,565**
Non controlling interest	150	(2,533)
NET PROFIT ATTRIBUTABLE TO THE COMPANY	**90,769**	**65,032**
BASIC EARNINGS PER SHARE	**0.80**	**0.58**
DILUTED EARNINGS PER SHARE	**0.79**	**0.57**
GAAP NET PROFIT	**90,769**	**65,032**
Stock based compensation and amortization of acquired intangible assets	9,941	9,441
ADJUSTED NET PROFIT	**100,710**	**74,473**
ADJUSTED EARNINGS PER SHARE:		
Basic	**0.89**	**0.66**
Diluted	**0.88**	**0.65**

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Primary source exhibit T8. Tower Q2 2026 supporting slides, August 4, 2026, PDF page 8. Reported Q2 GAAP financials and reconciled adjusted EPS; dollars in thousands in the source image.

4. Local evidence and calibration packet

Q2 financial statements establish 460.079M revenue, 137.772M GP and 90.290M GAAP OP. The first modeled quarter uses 520M revenue guidance with estimated product mix and costs, rather than representing unreported margins as actuals. All six product GP paths are explicit. The 2028 endpoint is a conditional company business model; matching its dollar totals is calibration, not evidence that the execution risk has disappeared.

The dated primary sources support expansion and contracts, while the exact wafer allocation, ASP, billing realization and quarter-by-quarter conversion assumptions below remain expert/model inputs. The official Q2 call document is dated **August 4, 2026**.

2028 Target Business Model ($M)

Revenue	3,600
Cost of Revenue	1,970
Gross Profit	1,630
Gross Margin	***45.3%***
Op-Ex	250
Operating Profit	1,380
Operating Margin	***38.3%***
Net Profit	1,200
Net Profit Margin	***33.3%***

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Primary source exhibit T9. Tower Q2 2026 supporting slides, August 4, 2026, PDF page 5. Conditional 2028 management business model used to calibrate the retained reference. Net profit 1,200 is distinct from operating profit 1,380; USD millions.

5. Summing thesis score

Category	Points / maximum	Evidence
Core Thesis	17/20	Contracted customer demand, qualified capacity and now volume laser-integrated PIC shipments.
Cycle	8/10	Revenue and margins are inflecting with capacity coming online.
TAM	9/10	Pluggable, NPO, coherent and heterogeneous integration extend optical content.
Catalysts	5/5	2026 capacity ramp, 2027 NPO shipments and Japan Track 1 qualification.
Macro	8/10	AI interconnect expansion; technology routes and global trade remain uncertain.
Alpha	2/5	Strong thesis and financial targets are already public.
Risks	-4	Capacity funding/execution, wafer prices, concentrated customers and geopolitics.

Sum: 45/60 × 100 = 75.000000%, conventionally rounded to 75%. This advisory thesis score does not replace any portfolio Base override. The separate risk classes below total the displayed deduction; holding status and classification do not waive that deduction.

October 3 evidence translation and probability ledger

One source chain, distinct outcomes. The P Equity Research / tuolaji item was published October 2, 2026; its interview date and expert identity are undisclosed, and the supplied document describes an AI translation. Public evidence independently supports foundry expansion and ST's entry. Exact allocation, wafer pricing and yield claims remain attributed hypotheses. Probability weights attach to bounded operating outcomes, not the probability that Tower's whole business works.

The capacity outcome has **25%** probability, competitive pricing without that acceleration has **20%**, and the unchanged reference has **55%**. These three states are mutually exclusive and sum to 100%. The temporary yield stress has a separate **20%** weight on Q4 2026 / Q1 2027 manufacturing costs only. It is not an independent confirmation of the other claims, and it lies outside the active Q4 2027-Q3 2028 earnings window. The specific SiGe allocation hypothesis carries **30%**, with zero incremental economics after subtracting the existing path.

The prior report's 75% execution-confidence label for the 2028 management reference is retained only as a qualitative evidence label. It is not another multiplier on these revenues, operating profit or the final price target. Artifact Base 75 and the protected Excel Base 99 likewise do not weight the price target.

Capacity pools, shipments and full-reference subtraction

The conditional path uses 360,000 actual **200mm** wafers annually at USD 3,500 and 55,000 / 100,000 actual **300mm** wafers in 2027 / 2028 at USD 9,000, with **90% billed realization**. These are distinct wafer-size pools; no 2.25x diameter conversion is applied to 300mm counts before using their 300mm ASP. Billed realization is an analyst conversion proxy, not reported die yield or the expert's claimed booking percentage. The 2027 Japan quarter ramp is 5,000 / 10,000 / 17,500 / 22,500 wafers and 2028 is 25,000 per quarter.

The resulting 2027 conditional SiPho sales are USD 1,579.5M, versus the full existing 1,400M reference; the 2028 conditional sales are USD 1,944M, versus the full 1,989.5M reference. The model subtracts all existing SiPho revenue, including the public USD 1.3B 2027 contract book, NPO, NewPhotonics and Japan growth. The contract book covers all SiPho, not a separately verified 200mm-only book. Adding Japan wafer sales to that entire contract book would duplicate part of the forecast.

This is a differentiated timing and realization scenario, **not a hard maximum capacity limit**. Existing Japan 300mm production and added output during 2027 precede full Track 1 readiness in Q4 2027. The new 2029 fab is not needed to justify every 2027 wafer. In 2029 onward, the capacity branch returns to the retained reference as the existing Track 2 path takes over; it does not impose an 8-inch or 12-inch capacity cap on the outer years.

company's total available capacity ceiling is 10,000 wafers per month, of which 9,000 will go to silicon-photonics production. During the ramp-up phase, full-year 2027 12-inch silicon-photonics capacity is expected to be about 50,000–60,000 wafers.

Q10. Given such large-scale capital investment, how does the company assess the return on investment and strategic necessity? What does this mean for solidifying its market position and addressing future risk?

This large-scale investment is based on considerations of strategic necessity and a high return on investment. Without this expansion, the company would be unable to meet customer demand, leading to customer loss. Through bold investment in its 8-inch line, the company has successfully locked in four major customers through 2028, effectively blocking competitors from entering this market segment.

Expert source exhibit E12. P Equity Research / tuolaji, October 2, 2026, PDF page 12. Attributed 2027 Japan 300mm pool and customer-demand discussion. The model uses actual 300mm wafers, not 200mm equivalents; quantities and realization are not issuer guidance. Anonymous expert / AI translation; claims are not issuer-confirmed. White margins trimmed; source text unchanged. Original exhibit, panel 1 of 2; read from top to bottom.

Although Sirui does not work with the company, its market share is relatively small, mainly supplying customers like Accelink and AOI, with monthly volume of only a few thousand wafers. In terms of ROI, although capital expenditure is large, the silicon-photonics business is highly profitable. In addition, building the new 12-inch fab is also meant to resolve the company's long-standing shortage of 12-inch capacity. Currently, the company's 65nm RF SOI products for mobile front-end devices are made by leasing fab space from STMicroelectronics in Italy; building its own new fab would eliminate potential future supply-chain risk and ensure the company controls its own capacity.

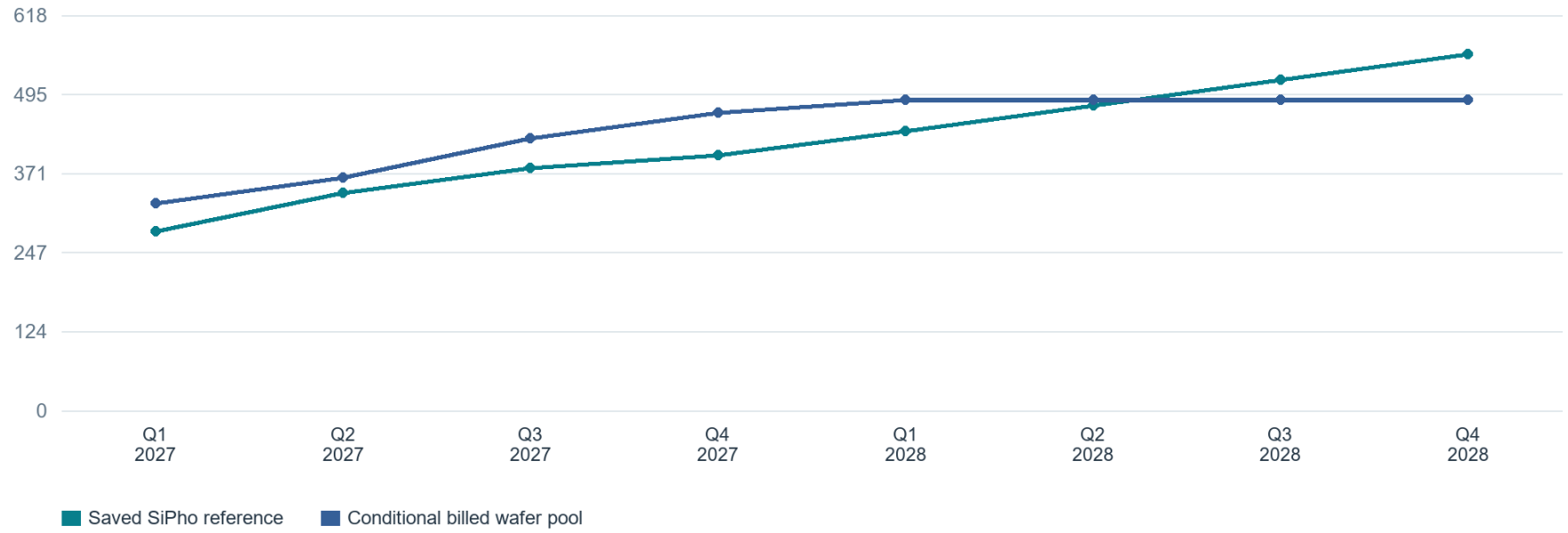
Q11. Of the 10,000 wafers of total 12-inch capacity, why is only 9,000 being converted to silicon photonics? What is the remaining 1,000 used for?

Of the 10,000 wafers per month of total 12-inch capacity, 1,000 wafers are reserved to meet sensor demand.

Expert source exhibit E12. P Equity Research / tuolaji, October 2, 2026, PDF page 12. Attributed 2027 Japan 300mm pool and customer-demand discussion. The model uses actual 300mm wafers, not 200mm equivalents; quantities and realization are not issuer guidance. Anonymous expert / AI translation; claims are not issuer-confirmed. White margins trimmed; source text unchanged. Original exhibit, panel 2 of 2; read from top to bottom.

Wafer-based conditional pool versus existing SiPho

USD millions / quarter; unweighted



Model figure. Wafer-based conditional pool versus existing SiPho. USD millions / quarter; unweighted. Root-frozen inputs, research October 3, 2026; conditional scenarios are not issuer guidance.

Modeled quarter	200mm wafers	300mm wafers	Reference SiPho	Conditional billed pool	Unweighted difference
Q1 2027	90,000	5,000	280.000	324.000	44.000
Q2 2027	90,000	10,000	340.000	364.500	24.500
Q3 2027	90,000	17,500	380.000	425.250	45.250
Q4 2027	90,000	22,500	400.000	465.750	65.750
Q1 2028	90,000	25,000	437.690	486.000	48.310
Q2 2028	90,000	25,000	477.480	486.000	8.520
Q3 2028	90,000	25,000	517.270	486.000	-31.270
Q4 2028	90,000	25,000	557.060	486.000	-71.060

Conditional pool = [(200mm wafers x USD 3,500) + (300mm wafers x USD 9,000)] x 90% / 1,000,000. Incremental branch revenue = this pool minus the full saved SiPho reference. The branch can produce negative differences in later 2028 quarters; the model does not keep only favorable quarters.

Primary-source excerpts, paraphrased: the May 13 contracts announcement reports USD 1.3B of 2027 SiPho delivery commitments and USD 290M of prepayments. The official August 4 call, PDF pages 3-4 and 8-9, separates capacity becoming qualified and shipped from booked demand and discusses the 2028 run-rate qualification. The July 14 FAQ, PDF pages 2-3, dates full Track 1 readiness to Q4 2027 and places Track 2 contribution in 2029. These do not establish an exclusive 200mm contract denominator, exact customer allocation or a hard 2027 shipment cap.

2028 Target Business Model ($M)

	FY 2025	Feb' 2026 Model		Updated Aug' 2026 Model	Incremental vs. Feb' 2026 Model	
					$M of Increase	% of Incremental Revenue
Revenue	1,566	2,840	1.3X	3,600	760	
Gross Profit	364	1,120	1.5X	1,630	510	67%
Operating Profit	194	900	1.5X	1,380	480	63%
Net Profit	220	750	1.6X	1,200	450	59%

** Model assumes 85% utilization across all of Tower's foundry capacity.*

*** Including revenue and margins from the $920M SiGe & SiPho capacity and capability Cap-Ex investments announced prior to Q3'2026 and Japan Track I / Arai related added SiPho Cap-Ex announced on July 14, 2026.*

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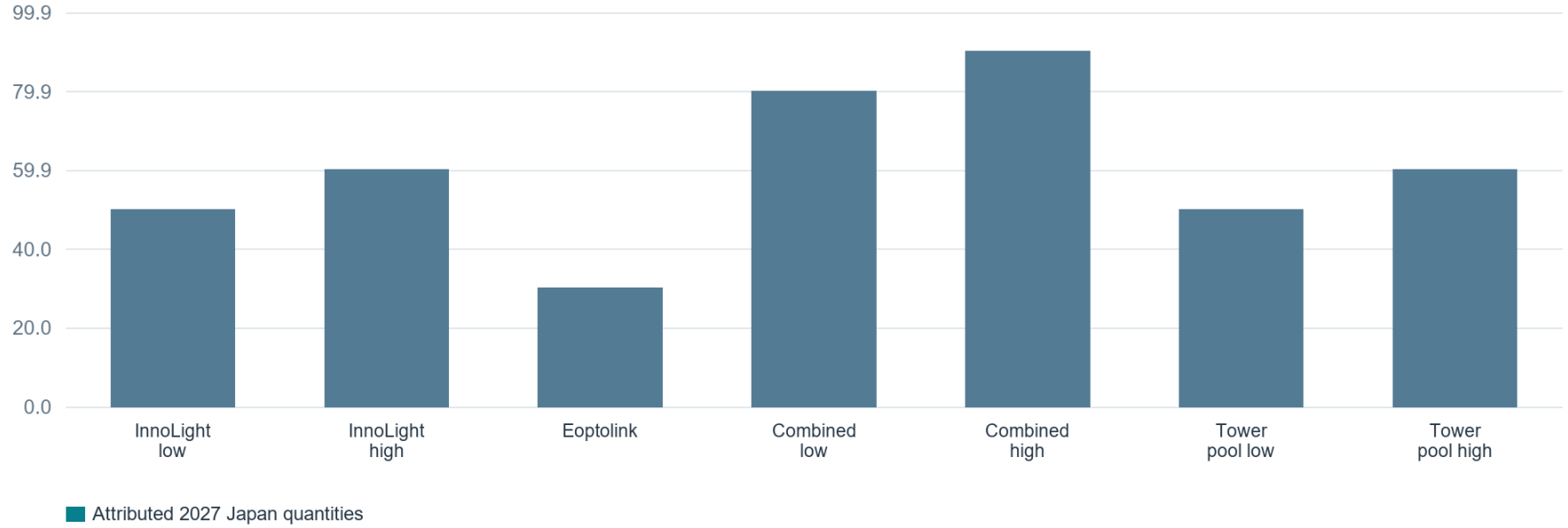
Primary source exhibit T14. Tower Q2 supporting slides, August 4, 2026, PDF page 4. The management 2028 reference includes earlier SiGe / SiPho investments and Japan Track 1, with an 85% utilization assumption. It is not an expert-call additive revenue layer.

Customer requests are not additive allocations

The expert claims that four customers account for more than 90% of the 2027 200mm book, without disclosing a complete customer-by-customer allocation. Its Japan examples use requests and demand forecasts that can exceed one available pool. The following check deliberately compares demand with the same denominator; it does not treat each request as an extra factory or a confirmed sale.

Requests versus the shared Japan allocation budget

Thousands of 300 mm wafers / year; expert PDF page 21



Model figure. Requests versus the shared Japan allocation budget. Thousands of 300mm wafers / year; expert PDF page 21. Root-frozen inputs, research October 3, 2026; conditional scenarios are not issuer guidance.

Expert statement	2027 quantity	Definition / overlap check
InnoLight demand	50,000-60,000 300mm wafers	PDF page 21; demand, not separately confirmed allocation
Eoptolink demand	About 30,000 300mm wafers	PDF page 21; competes for the same Japan pool
Combined demand arithmetic	80,000-90,000 300mm wafers	Sum of the two requests, not booked Tower revenue
Tower pool in that comparison	50,000-60,000 300mm wafers	One denominator; source says priority InnoLight then Eoptolink then Xihe
Xihe 200mm agreement vs request	36,000 versus more than 54,000 wafers	PDF page 19; separate 200mm pool, not additive to Japan 300mm demand
InnoLight other source descriptions	53,000 annual versus 7,500/month from H2	PDF pages 9 and 15; timing/scope conflict, not reconciled sales guidance

Q31. Given ST's competitive advantages in capacity and pricing, will its 2027 silicon-photonics revenue be comparable to the company's? How will the company handle share allocation with core customers like InnoLight and Eoptolink?

The revenue comparison depends on how much order share ST ultimately wins away from the company. Take InnoLight as an example: its total demand is about 50,000–60,000 wafers; combined with Eoptolink's roughly 30,000 wafers, total market demand is close to 90,000–100,000 wafers. If the company reaches 60,000 wafers of capacity in 2027, even if InnoLight shifts some orders elsewhere, most of its business is expected to remain with the company. This is because the company has close R&D collaboration with InnoLight across product generations; if InnoLight significantly cut its order volume, it could hurt its own R&D progress, so starting from scratch with ST is not its best option.

Expert source exhibit E21. P Equity Research / tuolaji, October 2, 2026, PDF page 21. Competing Japan customer demand and allocation priority; the 50-60k pool cannot be sold to every requested customer quantity simultaneously. Anonymous expert / AI translation; claims are not issuer-confirmed. White margins trimmed; source text unchanged. Original exhibit, panel 1 of 2; read from top to bottom.

The company's strategy is therefore clear: prioritize fully meeting InnoLight's needs. Within its 50,000–60,000-wafer capacity plan, the company will first secure InnoLight's orders, with remainingcapacity allocated next to Eoptolink, then Xihe. By locking in InnoLight — a core customer with more than half the market — the company can secure its market position, with other smaller customers being relatively secondary. It is certain that some share will shift to ST, but the exact amount is still being negotiated.

Q32. Besides STMicroelectronics, who are the other competitors in the 12-inch silicon-photonics space? What is their development status and competitiveness?

Apart from ST, two additional competitors are UMC and Xinyang. UMC collaborates with SilTerra China on 12-inch product development. SilTerra China is not our current customer and remains small-scale, but if UMC and SilTerra's partnership progresses smoothly, they may

Expert source exhibit E21. P Equity Research / tuolaji, October 2, 2026, PDF page 21. Competing Japan customer demand and allocation priority; the 50-60k pool cannot be sold to every requested customer quantity simultaneously. Anonymous expert / AI translation; claims are not issuer-confirmed. White margins trimmed; source text unchanged. Original exhibit, panel 2 of 2; read from top to bottom.

Pricing and scope inconsistencies

The source's USD 3,500 x 2.25 diameter-area calculation equals USD 7,875, not USD 10,000. USD 8,000 per 300mm wafer is USD 3,555.56 per 200mm-equivalent area, about 1.59% above USD 3,500, before usable-die and process differences. A 20% discount exists only relative to a USD 10,000 same-size quote. The source also conflicts between monthly and annual ST volumes. Those claims are not treated as a verified 12x capacity increase or a like-for-like universal 20% Tower haircut.

The capacity branch selects USD 9,000 as a bounded model ASP between attributed USD 8,000 / 10,000 quotes. The separate competitive-pricing branch assumes a 10% concession on 25% of reference SiPho revenue, phased to half intensity in Q4 2027, full intensity in 2028 and 75% / 50% / 25% / zero during 2029. That 25% pool is an assumed unprotected/new-300mm exposure, not a disclosed contract cohort. It does not impose a blanket haircut on fixed-price 200mm commitments. Because it is mutually exclusive with the capacity branch, the model does not charge the selected USD 9,000 capacity price and a second concession on the same outcome.

Costs, yield and accounting

Capacity-driven incremental revenue carries **45% production cost + 7% ramp / rework cost + 3% manufacturing depreciation / SBC burden = 55% marginal COGS**, plus the saved quarterly opex/revenue ratio. These are low-confidence cost proxies. The same marginal 55% applies when conditional revenue falls below reference: that allows some ramp and depreciation cost to decline. This is an explicit cost-flexibility assumption, not a demonstrated fixed-cost reduction, depreciation add-back or grant income.

The pricing branch retains production cost on the same units, so its revenue reduction passes into GP; only the saved opex ratio provides limited expense relief. The temporary Fab 9 yield stress applies to one-sixth of the reference SiPho cost pool, half of those equivalent lots, with 75% / 85% stressed yield in Q4 2026 / Q1 2027 against a 90% target. Incremental COGS = reference SiPho COGS x (5,000 / 30,000) x 50% x (90% / stressed yield - 1). It does not impose a company-wide yield number or an additional lost-sales multiplier.

Q14. Regarding the yield issues that arose during the process transfer at Fab 9, what is the current progress in resolving them, and what are the future yield-improvement targets?

This issue is being gradually resolved. Process transfers typically take about a year to stabilize; the silicon-photonics process transfer at Fab 9 began in Q4 2025, and the company expects related issues to be fully resolved by Q4 2026. At that point, the target is to raise Fab 9's yield to a level comparable to Fab 3's, i.e., above 90%. It should be noted that not all lots produced at Fab 9 have had serious problems — some products have achieved yields of 50%–60%.

By comparison, the process transfer at Fab 2 went much more smoothly. Fab 2 is one of the company's founding fabs, with extensive experience producing a wide range of product categories (such as sensors, BCD, RF SOI, CMOS, and discretes), and its engineering team has far more experience than Fab 9's. Fab 9 was acquired from Maxim

Expert source exhibit E14. P Equity Research / tuolaji, October 2, 2026, PDF page 14. Attributed Fab 9 transfer/yield recovery discussion. The report models a bounded temporary cost stress, not a verified company-wide yield failure. Anonymous expert / AI translation; claims are not issuer-confirmed. White margins trimmed; source text unchanged. Original exhibit, panel 1 of 2; read from top to bottom.

Integrated and previously only produced its own products, soit has relatively less experience handling complex process transfers. Thanks to Fab 2's stable performance, customers like InnoLight and Eoptolink now tend to place new products at Fab 2, only accepting Fab 9 production when there is no other option.

Q15. For 12-inch wafer demand in 2027–2028, especially in NPO-related applications, what is the current market-demand forecast?

According to forecasts, total market demand for 12-inch wafer NPO over the 2027–2028 delivery period breaks down as follows: NVIDIA is expected to need 25 million 3.2T units for its NVL576 platform; Amazon is expected to need 17 million 6.4T units for its Trainium4 platform; and other smaller CSP customers' combined demand is about 9 million units. In addition, since demand from Google and Meta has not yet fully materialized, more than 10 million additional units of demand may be added in the future.

Expert source exhibit E14. P Equity Research / tuolaji, October 2, 2026, PDF page 14. Attributed Fab 9 transfer/yield recovery discussion. The report models a bounded temporary cost stress, not a verified company-wide yield failure. Anonymous expert / AI translation; claims are not issuer-confirmed. White margins trimmed; source text unchanged. Original exhibit, panel 2 of 2; read from top to bottom.

Signal-by-signal disposition

PE-TSEM-20261002-CAPACITY

Field	Root-selected treatment
Probability	25%
Weighted hypothesis	Differentiated 2027-28 shipment path from the expert wafer-capacity envelope.
Rationale	One credible publisher/expert chain, unidentified speaker, translation conflicts and uncertain wafer-start versus accepted-shipment scope.
Reference / overlap	Subtract the full existing SiPho forecast, including 2027 contracts, NPO / NewPhotonics and Japan growth. Exact InnoLight requests are not added separately.
Active-window incremental OP	8.510347 USD M; probability applied once
Evidence trigger	Raise on matched accepted wafer deliveries, customer allocation and realized ASP. Reduce on mismatched shipment scope, duplicated fabs, qualification delays or unconverted commitments.

PE-TSEM-20261002-PRICING

Field	Root-selected treatment
Probability	20%
Weighted hypothesis	Competitive pricing concession on unprotected / new 300mm business, without the acceleration outcome.
Rationale	ST entry has primary support; exact prices and equivalent product/process scope remain uncertain. Fixed-price 200mm contracts limit exposure.
Reference / overlap	Mutually exclusive with the 25% capacity branch; residual 55% reference. The capacity case already uses USD 9,000 instead of USD 10,000.
Active-window incremental OP	-7.584625 USD M; probability applied once
Evidence trigger	Raise on comparable contracted/quoted prices and verified repriceable exposure. Remove if contracts protect pricing or quote differences reflect product/process scope.

PE-TSEM-20261002-YIELD

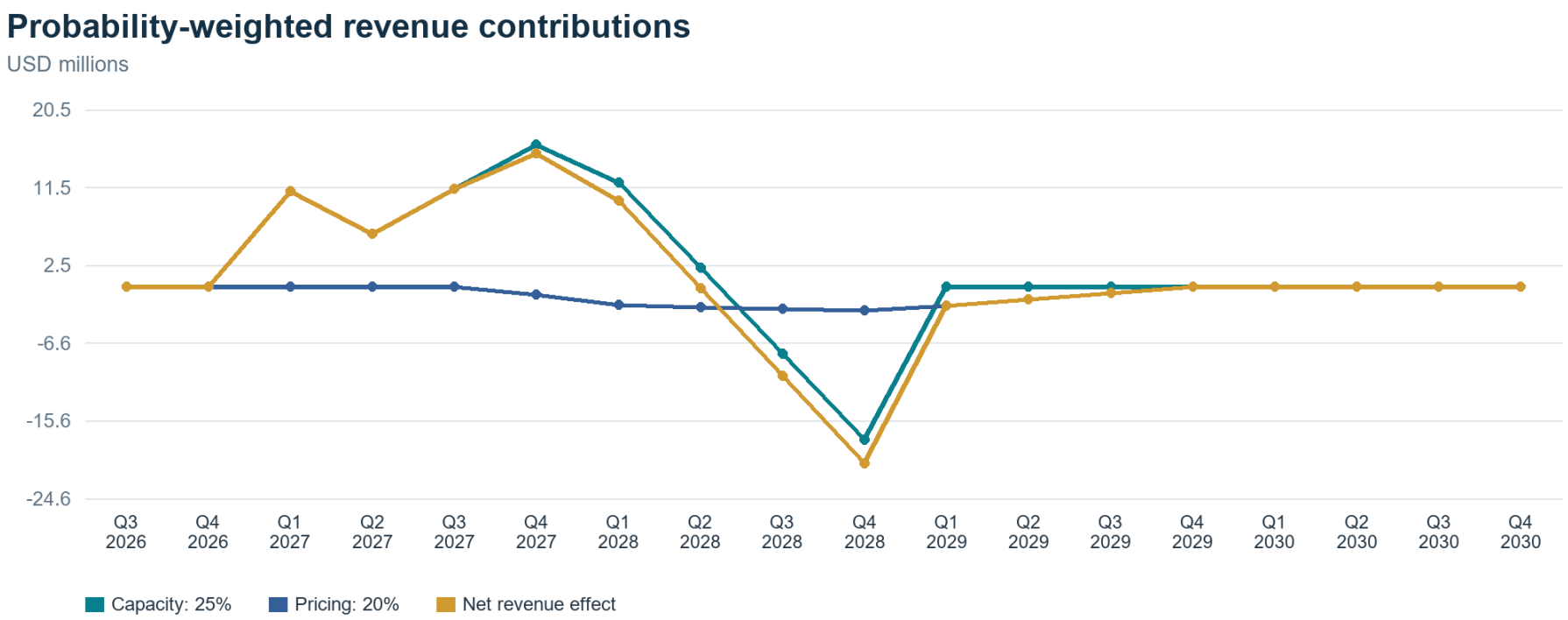
Field	Root-selected treatment
Probability	20%
Weighted hypothesis	Residual Fab 9 rework burden after Q3 2026, resolved by Q2 2027.
Rationale	The expert reports some affected lots at 50-60% and a recovery target above 90%; the selected case is a 20%-weighted partial-recovery stress.
Reference / overlap	Temporary manufacturing cost only. No Q3 guide cut or lost-sales double count. Entirely outside the active Q4 2027 earnings window.
Active-window incremental OP	0.000000 USD M; probability applied once
Evidence trigger	Remove on evidence of transfer/yield recovery; extend only if new evidence changes the timing.

PE-TSEM-20261002-SIGE

Field	Root-selected treatment
Probability	30%
Weighted hypothesis	Semtech 8,000/month and MACOM 3,500/month SiGe allocations by late 2027.
Rationale	Specific expert quantities with source and timing conflicts; Semtech primary evidence supports investment direction, not exact allocation.
Reference / overlap	Existing Q4 2027 SiGe revenue is already USD 100M. The attributed endpoints plus the existing USD 68M other-SiGe residual imply USD 99.8M/quarter. No separate revenue or profit is added.
Active-window incremental OP	0.000000 USD M; probability applied once
Evidence trigger	Customer-specific accepted wafers, realized ASP and other SiGe mix must exceed the already modeled path before recognizing incremental revenue.

SiGe overlap arithmetic: at the selected USD 2,400 per wafer proxy, Semtech 8,000/month plus MACOM 3,500/month implies USD 331.2M annual wafer sales. The current 3,750 + 2,500 monthly proxy implies USD 180M. Against the saved USD 248M annualized initial SiGe reference, the other-SiGe residual is USD 68M. Retaining that residual gives USD 399.2M annual / 99.8M quarterly at the endpoint, already covered by Q4 2027 SiGe of USD 100M. Saved FY2028 SiGe USD 412.821M requires about USD 81.621M of other SiGe, approximately 20% above the residual. These are attributed / model arithmetic, not new customer invoices.

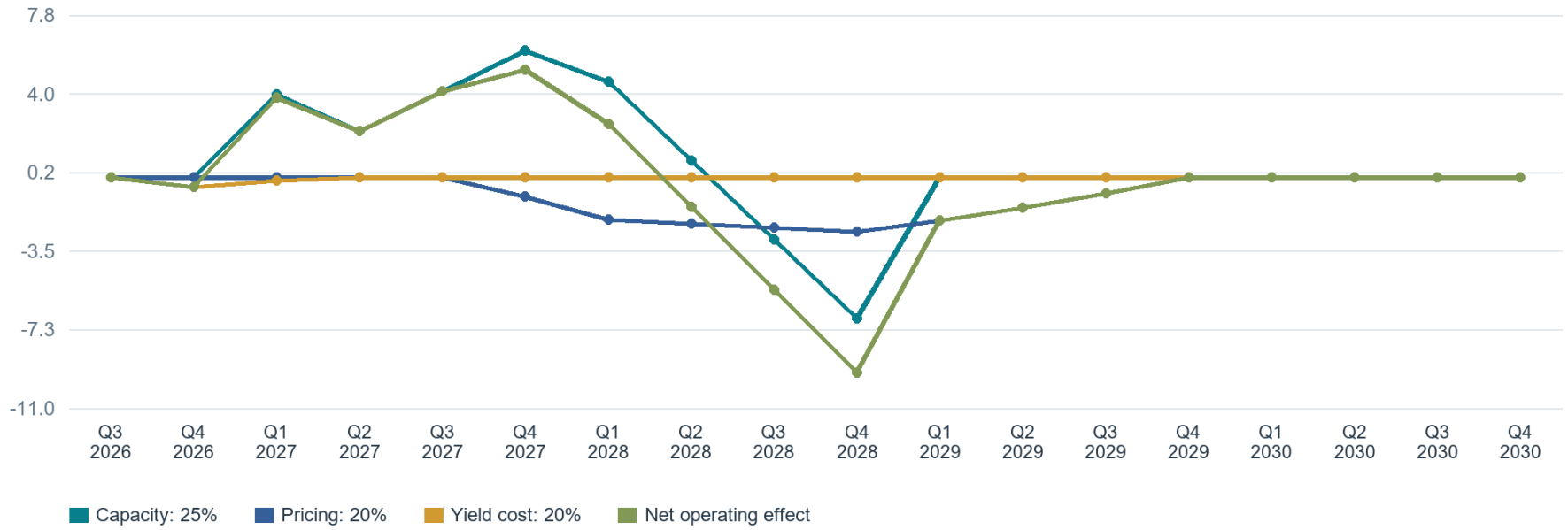
Quarterly probability-weighted revenue and OP effects



Model figure. Probability-weighted revenue contributions. USD millions. Root-frozen inputs, research October 3, 2026; conditional scenarios are not issuer guidance.

Probability-weighted operating profit contributions

USD millions



Model figure. Probability-weighted operating profit contributions. USD millions. Root-frozen inputs, research October 3, 2026; conditional scenarios are not issuer guidance.

Modeled quarter	Capacity delta revenue	Price delta revenue	Net delta revenue	Capacity delta OP	Price delta OP	Yield delta OP	Net delta OP
Q3 2026	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Q4 2026	0.0000	0.0000	0.0000	0.0000	0.0000	-0.4567	-0.4567
Q1 2027	11.0000	0.0000	11.0000	3.9622	0.0000	-0.1460	3.8162
Q2 2027	6.1250	0.0000	6.1250	2.2258	0.0000	0.0000	2.2258
Q3 2027	11.3125	0.0000	11.3125	4.1472	0.0000	0.0000	4.1472
Q4 2027	16.4375	-1.0000	15.4375	6.0786	-0.9198	0.0000	5.1588
Q1 2028	12.0775	-2.1884	9.8891	4.5962	-2.0365	0.0000	2.5597
Q2 2028	2.1300	-2.3874	-0.2574	0.8106	-2.2216	0.0000	-1.4110

Modeled quarter	Capacity delta revenue	Price delta revenue	Net delta revenue	Capacity delta OP	Price delta OP	Yield delta OP	Net delta OP
Q3 2028	-7.8175	-2.5863	-10.4038	-2.9750	-2.4067	0.0000	-5.3817
Q4 2028	-17.7650	-2.7853	-20.5503	-6.7606	-2.5919	0.0000	-9.3524
Q1 2029	0.0000	-2.2125	-2.2125	0.0000	-2.0612	0.0000	-2.0612
Q2 2029	0.0000	-1.5500	-1.5500	0.0000	-1.4448	0.0000	-1.4448
Q3 2029	0.0000	-0.8125	-0.8125	0.0000	-0.7577	0.0000	-0.7577
Q4 2029	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Q1 2030	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Q2 2030	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Q3 2030	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Q4 2030	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Unweighted scenario cost arithmetic: probability is applied after this table

Capacity branch

Quarter	Delta revenue	Delta COGS	Delta GP	Delta opex	Delta OP
Q3 2026	0.0000	0.0000	0.0000	0.0000	0.0000
Q4 2026	0.0000	0.0000	0.0000	0.0000	0.0000
Q1 2027	44.0000	24.2000	19.8000	3.9512	15.8488
Q2 2027	24.5000	13.4750	11.0250	2.1217	8.9033
Q3 2027	45.2500	24.8875	20.3625	3.7738	16.5886
Q4 2027	65.7500	36.1625	29.5875	5.2731	24.3143
Q1 2028	48.3100	26.5705	21.7395	3.3549	18.3847
Q2 2028	8.5200	4.6860	3.8340	0.5917	3.2423
Q3 2028	-31.2700	-17.1985	-14.0715	-2.1715	-11.9000
Q4 2028	-71.0600	-39.0830	-31.9770	-4.9347	-27.0423
Q1 2029	0.0000	0.0000	0.0000	0.0000	0.0000
Q2 2029	0.0000	0.0000	0.0000	0.0000	0.0000
Q3 2029	0.0000	0.0000	0.0000	0.0000	0.0000
Q4 2029	0.0000	0.0000	0.0000	0.0000	0.0000
Q1 2030	0.0000	0.0000	0.0000	0.0000	0.0000
Q2 2030	0.0000	0.0000	0.0000	0.0000	0.0000
Q3 2030	0.0000	0.0000	0.0000	0.0000	0.0000
Q4 2030	0.0000	0.0000	0.0000	0.0000	0.0000

Competitive-pricing branch

Quarter	Delta revenue	Delta COGS	Delta GP	Delta opex	Delta OP
Q3 2026	0.0000	0.0000	0.0000	0.0000	0.0000
Q4 2026	0.0000	0.0000	0.0000	0.0000	0.0000
Q1 2027	0.0000	0.0000	0.0000	0.0000	0.0000
Q2 2027	0.0000	0.0000	0.0000	0.0000	0.0000
Q3 2027	0.0000	0.0000	0.0000	0.0000	0.0000
Q4 2027	-5.0000	0.0000	-5.0000	-0.4010	-4.5990
Q1 2028	-10.9422	0.0000	-10.9422	-0.7599	-10.1824
Q2 2028	-11.9370	0.0000	-11.9370	-0.8290	-11.1080
Q3 2028	-12.9317	0.0000	-12.9317	-0.8980	-12.0337
Q4 2028	-13.9265	0.0000	-13.9265	-0.9671	-12.9594
Q1 2029	-11.0625	0.0000	-11.0625	-0.7567	-10.3058
Q2 2029	-7.7500	0.0000	-7.7500	-0.5262	-7.2238
Q3 2029	-4.0625	0.0000	-4.0625	-0.2742	-3.7883
Q4 2029	0.0000	0.0000	0.0000	0.0000	0.0000
Q1 2030	0.0000	0.0000	0.0000	0.0000	0.0000
Q2 2030	0.0000	0.0000	0.0000	0.0000	0.0000
Q3 2030	0.0000	0.0000	0.0000	0.0000	0.0000
Q4 2030	0.0000	0.0000	0.0000	0.0000	0.0000

Temporary yield-cost branch

Quarter	Delta revenue	Delta COGS	Delta GP	Delta opex	Delta OP
Q3 2026	0.0000	0.0000	0.0000	0.0000	0.0000
Q4 2026	0.0000	2.2833	-2.2833	0.0000	-2.2833
Q1 2027	0.0000	0.7302	-0.7302	0.0000	-0.7302
Q2 2027	0.0000	0.0000	0.0000	0.0000	0.0000
Q3 2027	0.0000	0.0000	0.0000	0.0000	0.0000
Q4 2027	0.0000	0.0000	0.0000	0.0000	0.0000
Q1 2028	0.0000	0.0000	0.0000	0.0000	0.0000
Q2 2028	0.0000	0.0000	0.0000	0.0000	0.0000
Q3 2028	0.0000	0.0000	0.0000	0.0000	0.0000
Q4 2028	0.0000	0.0000	0.0000	0.0000	0.0000
Q1 2029	0.0000	0.0000	0.0000	0.0000	0.0000
Q2 2029	0.0000	0.0000	0.0000	0.0000	0.0000
Q3 2029	0.0000	0.0000	0.0000	0.0000	0.0000
Q4 2029	0.0000	0.0000	0.0000	0.0000	0.0000
Q1 2030	0.0000	0.0000	0.0000	0.0000	0.0000
Q2 2030	0.0000	0.0000	0.0000	0.0000	0.0000
Q3 2030	0.0000	0.0000	0.0000	0.0000	0.0000
Q4 2030	0.0000	0.0000	0.0000	0.0000	0.0000

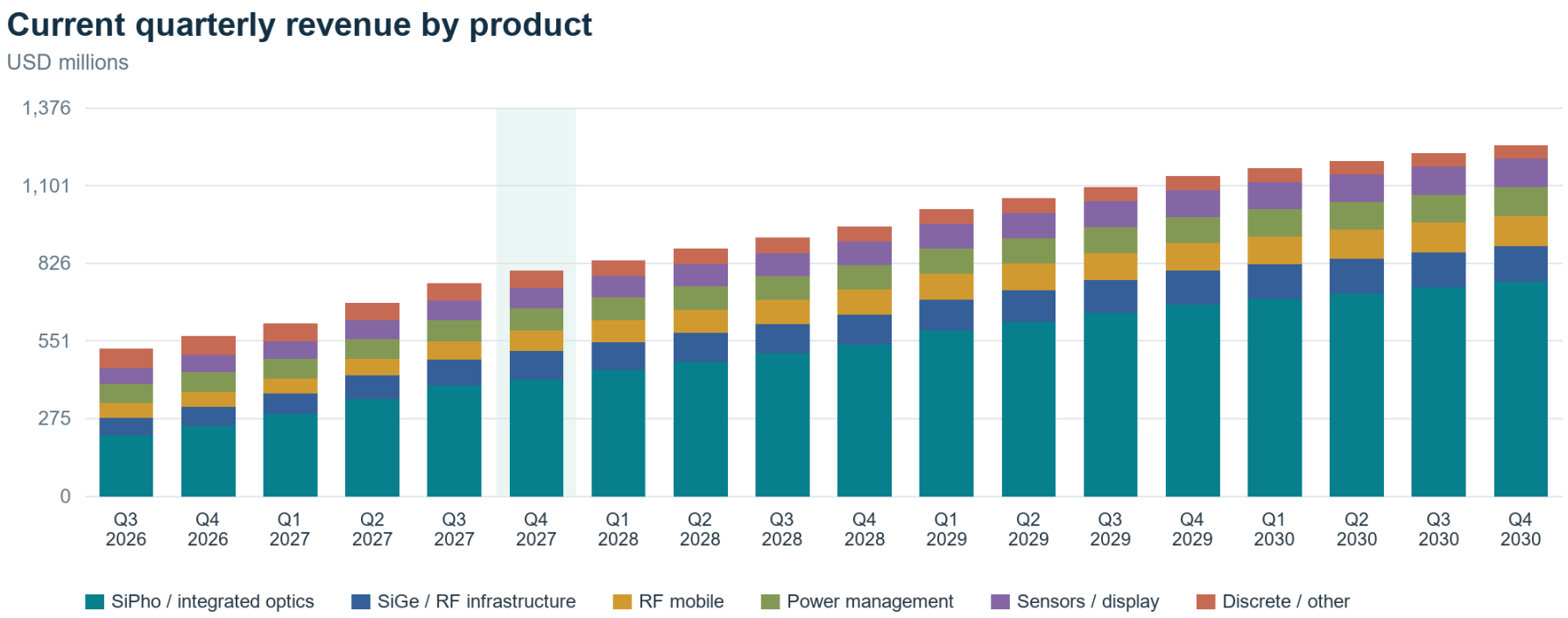
6. Revenue sources, capture and product margins

Calendar quarters and the existing 20x GAAP OP multiple. Six mutually exclusive product vectors derive consolidated GP and OP. The retained FY2028 reference product revenue sums to USD 3,600M, independently assigned product GMs sum to GP USD 1,630M, and explicit annual opex USD 250M gives OP USD 1,380M. Product allocations and margins are calibrated to this management endpoint; that calibration is not independent verification of the forecast. This reference matches the published model in dollars without treating rounded percentages as exact. Product-level GMs and quarterly timing remain analyst estimates. FY2027 reference SiPho revenue is USD 1,400M, including rather than adding USD 1,300M contracted sales. Track 1 supports FY2028; 2029-30 growth is a conditional realization of further capacity and demand, not quantified Track 2 guidance. NewPhotonics, integrated lasers, NPO/CPO and packaging are included in the SiPho vector, with no additive deal or packaging revenue. Shares use Q2 diluted 114.427M plus the retained analyst allowance of 0.4M per subsequent quarter, beginning at 114.827M in Q3. This is a forward dilution proxy, not management guidance. Management intent to fund construction internally is not a guarantee against financing or dilution. No cash, grant, customer-prepayment or net-interest amount is added to the OP price target.

Vector	Actual calibration / source	Forward demand and capture	Margin derivation / constraint
Silicon photonics	Q3 215M allocation, Q4 250M; Q2 disclosed annualized level above 680M.	Reference FY27 1.4B; FY28 about 1.9895B; FY29 2.54B; FY30 2.92B. The current weighted path is tabulated below. Demand expands through links, die area and integration. NewPhotonics, NPO/CPO and integrated lasers are included once.	Reference 43.5% initial GM rises into the 2028 ramp then normalizes to 50.2% at 2030 exit. The weighted current GM includes the new capacity, pricing and temporary yield effects. Process economics are estimates; manufacturing depreciation and SBC remain in the GAAP basis.
SiGe	62M initial allocation.	Customer bandwidth and analog performance requirements drive additional qualified wafer content. The quarterly path expands to 127M at 2030 exit, excluding SiPho invoices.	36% initial→mid/high 40s peak→44% exit. Assumed costs/yield and product pricing determine realization.
RF mobile	53M initial allocation.	Process migration and premium-device wins recover sales to 106M exit. Increased wafer starts are not identical to shipments or revenue before qualification and cycle time.	19.5%→around 32% peak→30.9% exit; loading and 300 mm migration improve absorption.
Power management	67M initial allocation.	BCD and differentiated power demand lift exit sales 102M; no full data-center power-market layer is added.	31.2%→around 37% peak→36% exit; modest process/mix improvement.

Vector	Actual calibration / source	Forward demand and capture	Margin derivation / constraint
Sensors/display	58M initial allocation.	Machine vision, inspection and other sensing demand raise exit sales 101M. Memory-fab demand is a downstream driver, not a guaranteed order for all sensor products.	27.2%→around 33% peak→32% exit.
Discrete/other	65M initial allocation.	Declining quarterly sales to 44M exit reflects a mature/commoditizing residual. The model does not apply optical growth universally.	15.5%→around 20% peak→18.6% exit; limited productivity benefit.

Confidence: reported historical totals are HIGH. Source-based forward extrapolations are MEDIUM only where the cited source sizes the relevant period; other product allocations, future capture and margin assumptions are LOW. A source for the historical anchor does not make an unsourced future factor management guidance. The detailed factors below are the current probability-weighted case, with uncertain assumptions exposed for review.



Model figure. Current quarterly revenue by product. USD millions. Root-frozen inputs, research October 3, 2026; conditional scenarios are not issuer guidance.

Modeled quarter	SiPho	SiGe	RF mobile	Power	Sensors	Other	Total
Q3 2026	215.000	62.000	53.000	67.000	58.000	65.000	520.000
Q4 2026	250.000	68.000	52.000	69.000	61.000	64.000	564.000
Q1 2027	291.000	74.000	52.000	70.000	63.000	60.000	610.000
Q2 2027	346.125	82.000	58.000	72.000	66.000	59.000	683.125
Q3 2027	391.312	92.000	66.000	75.000	70.000	58.000	752.312
Q4 2027	415.438	100.000	72.000	78.000	74.000	57.000	796.438
Q1 2028	447.579	99.475	77.590	79.580	75.601	54.711	834.537
Q2 2028	477.223	102.459	81.569	81.569	78.585	53.716	875.123
Q3 2028	506.866	104.449	85.548	83.559	81.569	52.722	914.714
Q4 2028	536.510	106.438	89.527	85.548	84.554	51.727	954.305
Q1 2029	587.788	110.000	92.000	88.000	87.000	51.000	1,015.788
Q2 2029	618.450	113.000	94.000	90.000	89.000	50.000	1,054.450
Q3 2029	649.188	116.000	96.000	92.000	91.000	49.000	1,093.188
Q4 2029	680.000	119.000	98.000	94.000	93.000	48.000	1,132.000
Q1 2030	700.000	121.000	100.000	96.000	95.000	47.000	1,159.000
Q2 2030	720.000	123.000	102.000	98.000	97.000	46.000	1,186.000
Q3 2030	740.000	125.000	104.000	100.000	99.000	45.000	1,213.000
Q4 2030	760.000	127.000	106.000	102.000	101.000	44.000	1,240.000

V1. Silicon photonics including integrated optics

Source anchors: T1/T2/T3/T5/T6, and the explicit October 3 signal ledger. Revenue = reference SiPho revenue + 25% x conditional capacity difference + 20% x conditional pricing difference. Product GP = reference GP + the same weighted branch GP differences - 20% x the temporary yield cost. Product margin = current GP / current revenue. Displayed demand / conversion factors divide the current revenue by the retained 215M anchor; they are model outputs, not independent additional multipliers.

Quarter	Revenue anchor	Current conversion factor	Current revenue	Current product GM	Current GP
Q3 2026	215.000000	1.000000	215.000	43.500%	93.525
Q4 2026	215.000000	1.162791	250.000	45.017%	112.543
Q1 2027	215.000000	1.353488	291.000	46.682%	135.844
Q2 2027	215.000000	1.609884	346.125	48.143%	166.636
Q3 2027	215.000000	1.820058	391.312	49.273%	192.811
Q4 2027	215.000000	1.932267	415.438	50.163%	208.397
Q1 2028	215.000000	2.081763	447.579	51.571%	230.823
Q2 2028	215.000000	2.219640	477.223	52.724%	251.611
Q3 2028	215.000000	2.357517	506.866	53.899%	273.195
Q4 2028	215.000000	2.495394	536.510	54.573%	292.789
Q1 2029	215.000000	2.733895	587.788	53.827%	316.388
Q2 2029	215.000000	2.876512	618.450	53.383%	330.150
Q3 2029	215.000000	3.019477	649.188	52.741%	342.387
Q4 2029	215.000000	3.162791	680.000	52.100%	354.280
Q1 2030	215.000000	3.255814	700.000	51.500%	360.500
Q2 2030	215.000000	3.348837	720.000	51.000%	367.200
Q3 2030	215.000000	3.441860	740.000	50.600%	374.440
Q4 2030	215.000000	3.534884	760.000	50.200%	381.520

V2. SiGe and other RF infrastructure

Sources: T1/2/4; SiGe applications and Japan capacity. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2026	62.000000	1.000000	62.000	36.000%	22.320

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q4 2026	62.000000	1.096774	68.000	37.500%	25.500
Q1 2027	62.000000	1.193548	74.000	39.000%	28.860
Q2 2027	62.000000	1.322581	82.000	40.500%	33.210
Q3 2027	62.000000	1.483871	92.000	42.000%	38.640
Q4 2027	62.000000	1.612903	100.000	43.500%	43.500
Q1 2028	62.000000	1.604435	99.475	45.295%	45.057
Q2 2028	62.000000	1.652568	102.459	46.295%	47.433
Q3 2028	62.000000	1.684657	104.449	46.795%	48.877
Q4 2028	62.000000	1.716746	106.438	47.295%	50.340
Q1 2029	62.000000	1.774194	110.000	46.800%	51.480
Q2 2029	62.000000	1.822581	113.000	46.400%	52.432
Q3 2029	62.000000	1.870968	116.000	45.900%	53.244
Q4 2029	62.000000	1.919355	119.000	45.400%	54.026
Q1 2030	62.000000	1.951613	121.000	45.000%	54.450
Q2 2030	62.000000	1.983871	123.000	44.600%	54.858
Q3 2030	62.000000	2.016129	125.000	44.300%	55.375
Q4 2030	62.000000	2.048387	127.000	44.000%	55.880

V3. RF Mobile

Sources: T1/2; RF SOI process transitions. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2026	53.000000	1.000000	53.000	19.500%	10.335
Q4 2026	53.000000	0.981132	52.000	21.200%	11.024

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q1 2027	53.000000	0.981132	52.000	22.800%	11.856
Q2 2027	53.000000	1.094340	58.000	24.400%	14.152
Q3 2027	53.000000	1.245283	66.000	26.000%	17.160
Q4 2027	53.000000	1.358491	72.000	27.800%	20.016
Q1 2028	53.000000	1.463972	77.590	30.295%	23.506
Q2 2028	53.000000	1.539047	81.569	31.295%	25.527
Q3 2028	53.000000	1.614123	85.548	31.795%	27.200
Q4 2028	53.000000	1.689198	89.527	32.295%	28.913
Q1 2029	53.000000	1.735849	92.000	32.000%	29.440
Q2 2029	53.000000	1.773585	94.000	31.800%	29.892
Q3 2029	53.000000	1.811321	96.000	31.600%	30.336
Q4 2029	53.000000	1.849057	98.000	31.400%	30.772
Q1 2030	53.000000	1.886792	100.000	31.200%	31.200
Q2 2030	53.000000	1.924528	102.000	31.100%	31.722
Q3 2030	53.000000	1.962264	104.000	31.000%	32.240
Q4 2030	53.000000	2.000000	106.000	30.900%	32.754

V4. Power management

Sources: T1/2; BCD/power adoption. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2026	67.000000	1.000000	67.000	31.200%	20.904
Q4 2026	67.000000	1.029851	69.000	32.000%	22.080
Q1 2027	67.000000	1.044776	70.000	33.000%	23.100

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q2 2027	67.000000	1.074627	72.000	34.000%	24.480
Q3 2027	67.000000	1.119403	75.000	35.000%	26.250
Q4 2027	67.000000	1.164179	78.000	35.500%	27.690
Q1 2028	67.000000	1.187761	79.580	36.295%	28.883
Q2 2028	67.000000	1.217455	81.569	36.795%	30.013
Q3 2028	67.000000	1.247149	83.559	37.295%	31.163
Q4 2028	67.000000	1.276843	85.548	37.495%	32.076
Q1 2029	67.000000	1.313433	88.000	37.100%	32.648
Q2 2029	67.000000	1.343284	90.000	36.900%	33.210
Q3 2029	67.000000	1.373134	92.000	36.700%	33.764
Q4 2029	67.000000	1.402985	94.000	36.500%	34.310
Q1 2030	67.000000	1.432836	96.000	36.300%	34.848
Q2 2030	67.000000	1.462687	98.000	36.200%	35.476
Q3 2030	67.000000	1.492537	100.000	36.100%	36.100
Q4 2030	67.000000	1.522388	102.000	36.000%	36.720

V5. Sensors and display

Sources: T1/2; image sensors and inspection demand. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2026	58.000000	1.000000	58.000	27.200%	15.776
Q4 2026	58.000000	1.051724	61.000	28.200%	17.202
Q1 2027	58.000000	1.086207	63.000	29.200%	18.396
Q2 2027	58.000000	1.137931	66.000	30.200%	19.932

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2027	58.000000	1.206897	70.000	31.000%	21.700
Q4 2027	58.000000	1.275862	74.000	31.500%	23.310
Q1 2028	58.000000	1.303465	75.601	32.295%	24.415
Q2 2028	58.000000	1.354918	78.585	32.795%	25.772
Q3 2028	58.000000	1.406371	81.569	33.295%	27.158
Q4 2028	58.000000	1.457823	84.554	33.495%	28.321
Q1 2029	58.000000	1.500000	87.000	33.100%	28.797
Q2 2029	58.000000	1.534483	89.000	32.900%	29.281
Q3 2029	58.000000	1.568966	91.000	32.700%	29.757
Q4 2029	58.000000	1.603448	93.000	32.500%	30.225
Q1 2030	58.000000	1.637931	95.000	32.300%	30.685
Q2 2030	58.000000	1.672414	97.000	32.200%	31.234
Q3 2030	58.000000	1.706897	99.000	32.100%	31.779
Q4 2030	58.000000	1.741379	101.000	32.000%	32.320

V6. Discrete and other

Sources: T1/2; mature discrete/other applications. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2026	65.000000	1.000000	65.000	15.500%	10.075
Q4 2026	65.000000	0.984615	64.000	16.200%	10.368
Q1 2027	65.000000	0.923077	60.000	17.000%	10.200
Q2 2027	65.000000	0.907692	59.000	17.700%	10.443
Q3 2027	65.000000	0.892308	58.000	18.300%	10.614

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q4 2027	65.000000	0.876923	57.000	18.700%	10.659
Q1 2028	65.000000	0.841711	54.711	19.295%	10.556
Q2 2028	65.000000	0.826408	53.716	19.595%	10.526
Q3 2028	65.000000	0.811104	52.722	19.795%	10.436
Q4 2028	65.000000	0.795800	51.727	19.995%	10.343
Q1 2029	65.000000	0.784615	51.000	19.600%	9.996
Q2 2029	65.000000	0.769231	50.000	19.400%	9.700
Q3 2029	65.000000	0.753846	49.000	19.200%	9.408
Q4 2029	65.000000	0.738462	48.000	19.000%	9.120
Q1 2030	65.000000	0.723077	47.000	18.900%	8.883
Q2 2030	65.000000	0.707692	46.000	18.800%	8.648
Q3 2030	65.000000	0.692308	45.000	18.700%	8.415
Q4 2030	65.000000	0.676923	44.000	18.600%	8.184

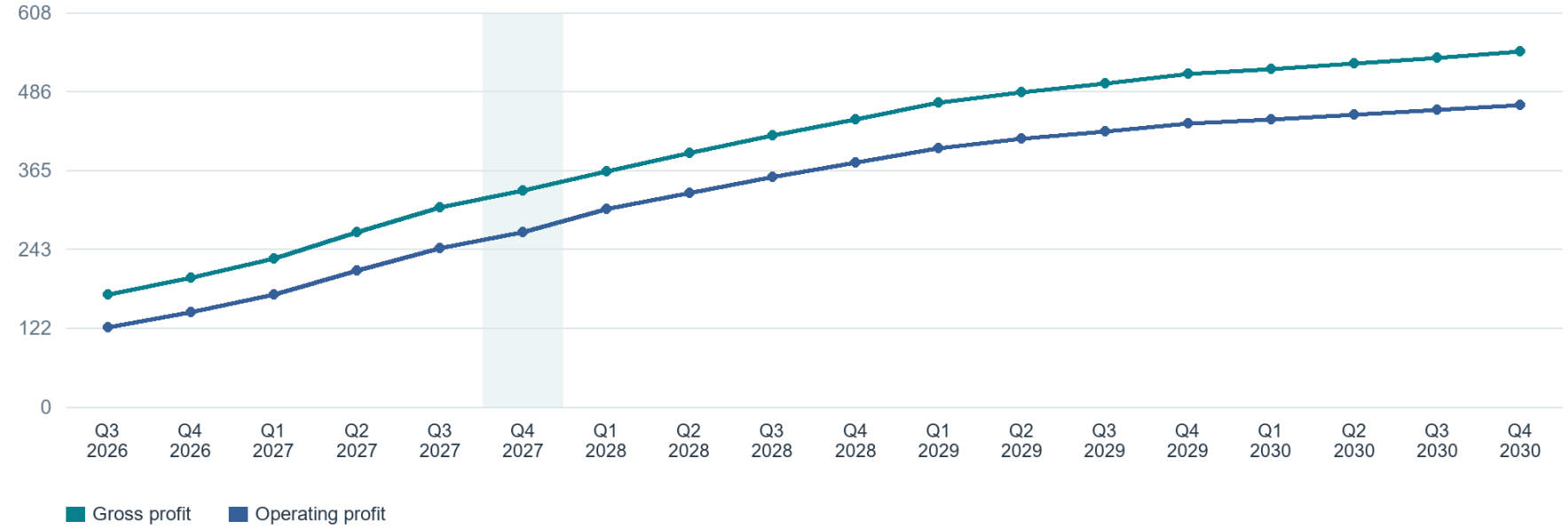
7. Operating-cost bridge and current quarterly model

Operating expenses are explicit and include GAAP compensation. The 2028 reference is USD 1,630M GP less USD 250M opex = USD 1,380M OP. The updated weighted schedule below applies the unchanged quarter-specific opex ratio to current revenue; all new manufacturing costs are charged through GP. Lower ratios reflect scale while reference dollar R&D grows. Manufacturing depreciation is not added back. Japan capex is a cash investment whose resulting depreciation belongs in product costs; customer advances fund capacity and also represent future delivery obligations.

Profit basis: consolidated GAAP operating profit after depreciation and stock compensation. All forward rows are modeled. Q3 2026 uses the revenue-guide midpoint, with estimated product mix and margins; no unreported quarter is represented as actual. Financial units: USD millions.

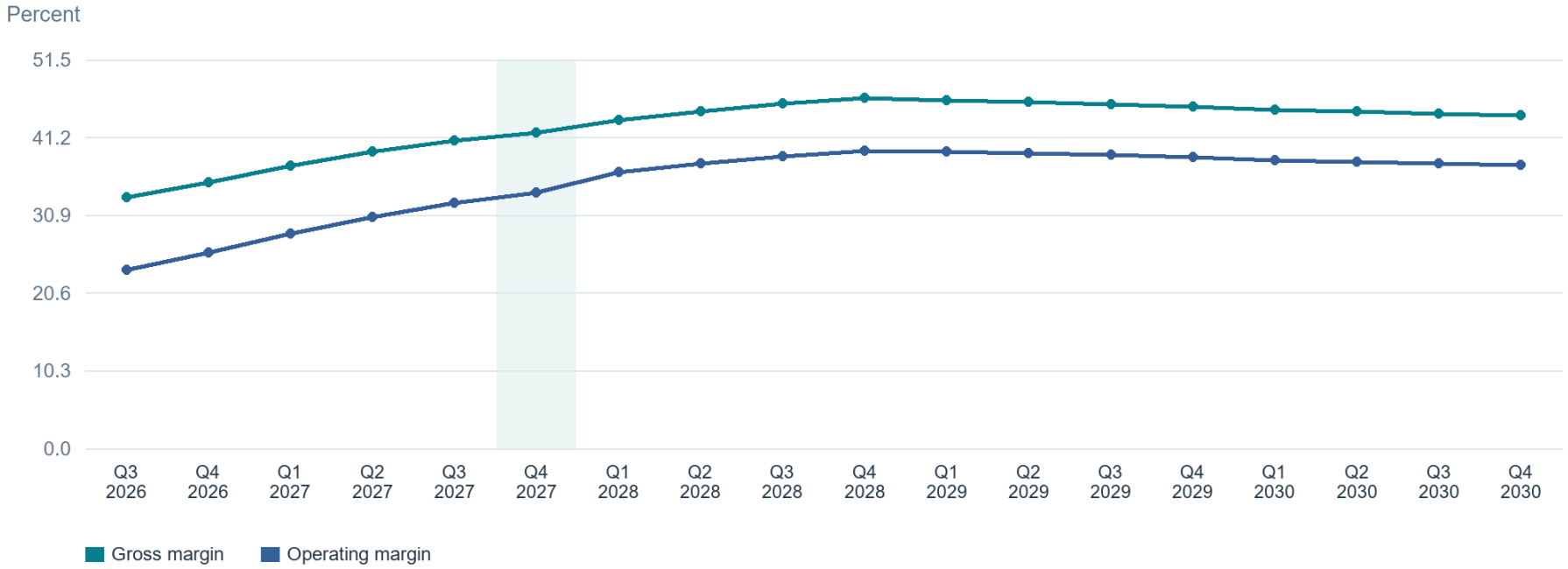
Current quarterly gross and operating profit

USD millions



Model figure. Current quarterly gross and operating profit. USD millions. Root-frozen inputs, research October 3, 2026; conditional scenarios are not issuer guidance.

Current consolidated margins



Model figure. Current consolidated margins. Percent. Root-frozen inputs, research October 3, 2026; conditional scenarios are not issuer guidance.

Modeled quarter	Revenue	GP	COGS	Opex rate	Opex	GAAP OP	OP margin
Q3 2026	520.000	172.935	347.065	9.600%	49.920	123.015	23.657%
Q4 2026	564.000	198.717	365.283	9.300%	52.452	146.265	25.934%
Q1 2027	610.000	228.256	381.744	8.980%	54.778	173.478	28.439%
Q2 2027	683.125	268.853	414.272	8.660%	59.159	209.695	30.696%
Q3 2027	752.312	307.175	445.138	8.340%	62.743	244.432	32.491%
Q4 2027	796.438	333.572	462.866	8.020%	63.874	269.698	33.863%
Q1 2028	834.537	363.241	471.296	6.944%	57.954	305.287	36.582%
Q2 2028	875.123	390.882	484.240	6.944%	60.772	330.110	37.722%
Q3 2028	914.714	418.029	496.684	6.944%	63.522	354.507	38.756%

Modeled quarter	Revenue	GP	COGS	Opex rate	Opex	GAAP OP	OP margin
Q4 2028	954.305	442.782	511.523	6.944%	66.271	376.511	39.454%
Q1 2029	1,015.788	468.749	547.039	6.840%	69.480	399.269	39.306%
Q2 2029	1,054.450	484.665	569.785	6.790%	71.597	413.068	39.174%
Q3 2029	1,093.188	498.897	594.291	6.750%	73.790	425.106	38.887%
Q4 2029	1,132.000	512.733	619.267	6.720%	76.070	436.663	38.574%
Q1 2030	1,159.000	520.566	638.434	6.690%	77.537	443.029	38.225%
Q2 2030	1,186.000	529.138	656.862	6.670%	79.106	450.032	37.945%
Q3 2030	1,213.000	538.349	674.651	6.650%	80.665	457.685	37.732%
Q4 2030	1,240.000	547.378	692.622	6.630%	82.212	465.166	37.513%

OP = sum of product GP - explicit operating expenses. GP = saved reference GP + probability-weighted incremental GP. Displayed numbers are rounded; the model uses unrounded values. The five non-SiPho product vectors are unchanged.

Reference GP plus incremental effects

Modeled quarter	Reference GP	Weighted delta GP	Current GP	Reference opex	Weighted delta opex	Current OP
Q3 2026	172.9350	0.0000	172.9350	49.9200	0.0000	123.0150
Q4 2026	199.1740	-0.4567	198.7173	52.4520	0.0000	146.2653
Q1 2027	223.4520	4.8040	228.2560	53.7902	0.9878	173.4780
Q2 2027	266.0970	2.7562	268.8533	58.6282	0.5304	209.6946
Q3 2027	302.0840	5.0906	307.1746	61.7994	0.9435	244.4318
Q4 2027	327.1750	6.3969	333.5719	62.6362	1.2381	269.6976
Q1 2028	359.9941	3.2464	363.2406	57.2672	0.6867	305.2866
Q2 2028	392.3109	-1.4289	390.8820	60.7903	-0.0179	330.1096
Q3 2028	424.1334	-6.1042	418.0292	64.2443	-0.7225	354.5074
Q4 2028	453.5615	-10.7795	442.7820	67.6983	-1.4271	376.5108
Q1 2029	470.9610	-2.2125	468.7485	69.6312	-0.1513	399.2686
Q2 2029	486.2150	-1.5500	484.6650	71.7024	-0.1052	413.0678
Q3 2029	499.7090	-0.8125	498.8965	73.8450	-0.0548	425.1063
Q4 2029	512.7330	0.0000	512.7330	76.0704	0.0000	436.6626
Q1 2030	520.5660	0.0000	520.5660	77.5371	0.0000	443.0289
Q2 2030	529.1380	0.0000	529.1380	79.1062	0.0000	450.0318
Q3 2030	538.3490	0.0000	538.3490	80.6645	0.0000	457.6845
Q4 2030	547.3780	0.0000	547.3780	82.2120	0.0000	465.1660

8. Multiple, shares and recognition convention

20×forward GAAP OP retains the framework valuation convention for differentiated foundry capacity. Competitive alternatives, capital requirements and pricing sensitivity constrain the multiple. The expanded TAM affects revenue/capture and mix rather than producing an automatic rerating.

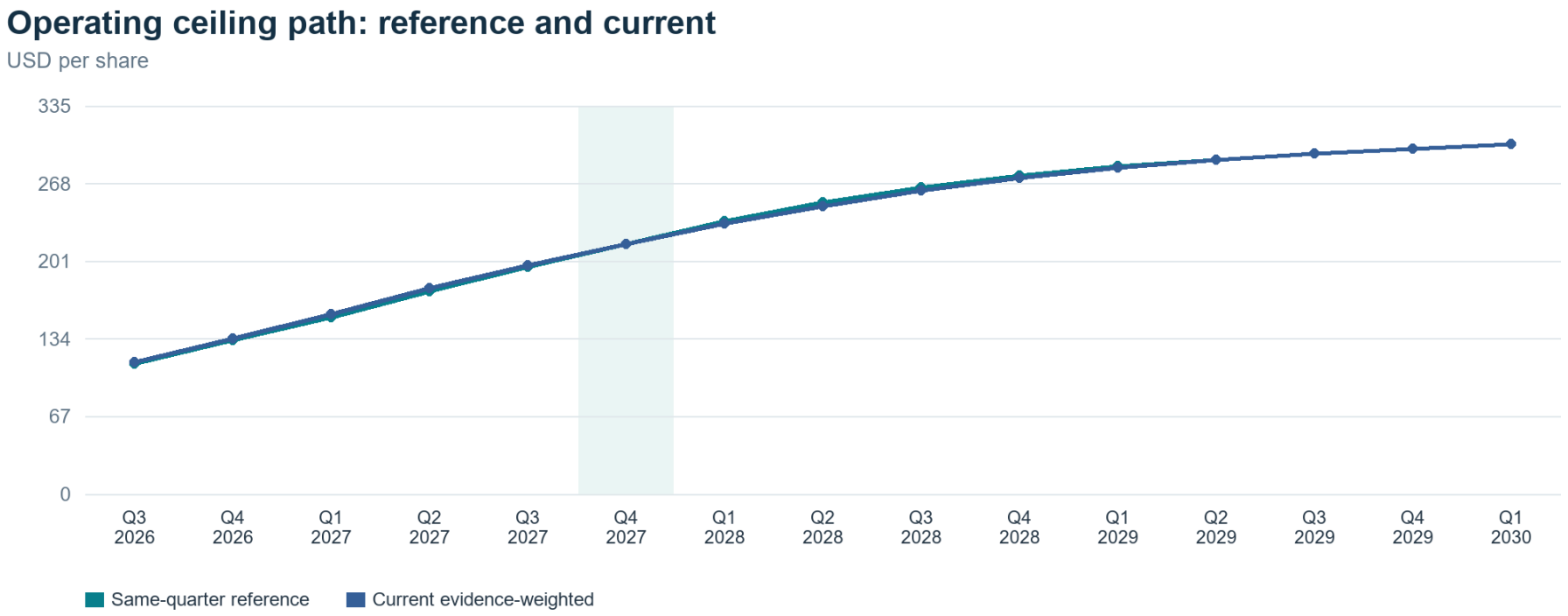
Q2 diluted 114.427M plus 0.4M each subsequent quarter gives 114.827M initially and 121.627M at 2030 exit. This is an exposed net-issuance allowance, not management guidance or a complete future award schedule. Management intends to finance Japan internally; that does not guarantee the absence of future debt or issuance. Approximately 3B net investment after 1B grants and 290M customer advances require careful cash-flow monitoring. Minority ownership and the Japan

restructuring can change parent economics even though the convention values consolidated OP.

The valuation denominator is the price target quarter's diluted-share proxy. It is not a historical EPS weighted-average assumption disguised as a fully known future capitalization. Unit factor: 1; this converts the model units to USD per share. No cash balance, enterprise-value adjustment or after-tax earnings substitution is added to the operating-profit price target.

9. Complete rolling four-quarter price target derivation

The active horizon is Q4 2027-Q3 2028. Both lines use the same 20x GAAP OP method and the same price target-quarter share proxy. The chart isolates the new operating evidence from the horizon roll.



Model figure. Operating price target path: reference and current. USD per share. Root-frozen inputs, research October 3, 2026; conditional scenarios are not issuer guidance.

price target quarter	Exact earnings window	Reference price target	Current F4Q GAAP OP	Multiple	Shares M	Current price target
Q3 2026	Q3 2026 to Q2 2027	112.668205	652.452919	20x	114.827000	113.641029

price target quarter	Exact earnings window	Reference price target	Current F4Q GAAP OP	Multiple	Shares M	Current price target
Q4 2026	Q4 2026 to Q3 2027	132.631623	773.869682	20x	115.227000	134.320894
Q1 2027	Q1 2027 to Q4 2027	152.551567	897.301936	20x	115.627000	155.206299
Q2 2027	Q2 2027 to Q1 2028	174.962577	1,029.110607	20x	116.027000	177.391574
Q3 2027	Q3 2027 to Q2 2028	195.671281	1,149.525631	20x	116.427000	197.467191
Q4 2027	Q4 2027 to Q3 2028	215.476825	1,259.601277	20x	116.827000	215.635303
Q1 2028	Q1 2028 to Q4 2028	235.440641	1,366.414493	20x	117.227000	233.122829
Q2 2028	Q2 2028 to Q1 2029	251.405350	1,460.396496	20x	117.627000	248.309741
Q3 2028	Q3 2028 to Q2 2029	264.616534	1,543.354692	20x	118.027000	261.525700
Q4 2028	Q4 2028 to Q3 2029	274.864625	1,613.953627	20x	118.427000	272.565146
Q1 2029	Q1 2029 to Q4 2029	282.489502	1,674.105424	20x	118.827000	281.771891
Q2 2029	Q2 2029 to Q1 2030	288.536674	1,717.865689	20x	119.227000	288.167225
Q3 2029	Q3 2029 to Q2 2030	293.510211	1,754.829644	20x	119.627000	293.383541
Q4 2029	Q4 2029 to Q3 2030	297.834287	1,787.407800	20x	120.027000	297.834287
Q1 2030	Q1 2030 to Q4 2030	301.578749	1,815.911200	20x	120.427000	301.578749

Complete quarter-by-quarter arithmetic

- **Q3 2026:** (123.015000 + 146.265333 + 173.477961 + 209.694625) x 20 / 114.827000 = **113.641029 USD/share**. Window: Q3 2026 to Q2 2027.
- **Q4 2026:** (146.265333 + 173.477961 + 209.694625 + 244.431762) x 20 / 115.227000 = **134.320894 USD/share**. Window: Q4 2026 to Q3 2027.
- **Q1 2027:** (173.477961 + 209.694625 + 244.431762 + 269.697588) x 20 / 115.627000 = **155.206299 USD/share**. Window: Q1 2027 to Q4 2027.
- **Q2 2027:** (209.694625 + 244.431762 + 269.697588 + 305.286632) x 20 / 116.027000 = **177.391574 USD/share**. Window: Q2 2027 to Q1 2028.
- **Q3 2027:** (244.431762 + 269.697588 + 305.286632 + 330.109650) x 20 / 116.427000 = **197.467191 USD/share**. Window: Q3 2027 to Q2 2028.
- **Q4 2027:** (269.697588 + 305.286632 + 330.109650 + 354.507408) x 20 / 116.827000 = **215.635303 USD/share**. Window: Q4 2027 to Q3 2028.
- **Q1 2028:** (305.286632 + 330.109650 + 354.507408 + 376.510804) x 20 / 117.227000 = **233.122829 USD/share**. Window: Q1 2028 to Q4 2028.
- **Q2 2028:** (330.109650 + 354.507408 + 376.510804 + 399.268635) x 20 / 117.627000 = **248.309741 USD/share**. Window: Q2 2028 to Q1 2029.
- **Q3 2028:** (354.507408 + 376.510804 + 399.268635 + 413.067845) x 20 / 118.027000 = **261.525700 USD/share**. Window: Q3 2028 to Q2 2029.

- **Q4 2028:** (376.510804 + 399.268635 + 413.067845 + 425.106344) x 20 / 118.427000 = **272.565146 USD/share**. Window: Q4 2028 to Q3 2029.
- **Q1 2029:** (399.268635 + 413.067845 + 425.106344 + 436.662600) x 20 / 118.827000 = **281.771891 USD/share**. Window: Q1 2029 to Q4 2029.
- **Q2 2029:** (413.067845 + 425.106344 + 436.662600 + 443.028900) x 20 / 119.227000 = **288.167225 USD/share**. Window: Q2 2029 to Q1 2030.
- **Q3 2029:** (425.106344 + 436.662600 + 443.028900 + 450.031800) x 20 / 119.627000 = **293.383541 USD/share**. Window: Q3 2029 to Q2 2030.
- **Q4 2029:** (436.662600 + 443.028900 + 450.031800 + 457.684500) x 20 / 120.027000 = **297.834287 USD/share**. Window: Q4 2029 to Q3 2030.
- **Q1 2030:** (443.028900 + 450.031800 + 457.684500 + 465.166000) x 20 / 120.427000 = **301.578749 USD/share**. Window: Q1 2030 to Q4 2030.

10. Annual trajectory and reference cross-check

These are calendar-year sums of the current quarterly path, not additional revenue or a different valuation method. The management-derived 2028 reference remains USD 3.6B revenue / 1.63B GP / 1.38B OP. The current expert-signal case is separately shown; successful full-year execution is distinct from reaching the same level only as a year-end run rate. Outer-year revenue is an analyst capacity/capture path, not verified management guidance.

Calendar year	Current revenue	Current GP	Current opex	Current OP	Current OP margin	Current SiPho
2027	2,841.875	1,137.856	240.554	897.302	31.574%	1,443.875
2028	3,578.678	1,614.934	248.519	1,366.414	38.182%	1,968.177
2029	4,295.425	1,965.043	290.938	1,674.105	38.974%	2,535.425
2030	4,798.000	2,135.431	319.520	1,815.911	37.847%	2,920.000

Annual product revenue and gross profit

2027 product totals

Product	Reference revenue	Current revenue	Current GP	Current GM
Silicon photonics including integrated optics	1,400.000	1,443.875	703.688	48.736%
SiGe and other RF infrastructure	348.000	348.000	144.210	41.440%
RF Mobile	248.000	248.000	63.184	25.477%
Power management	295.000	295.000	101.520	34.414%
Sensors and display	273.000	273.000	83.338	30.527%
Discrete and other	234.000	234.000	41.916	17.913%

2028 product totals

Product	Reference revenue	Current revenue	Current GP	Current GM
Silicon photonics including integrated optics	1,989.500	1,968.177	1,048.417	53.268%
SiGe and other RF infrastructure	412.821	412.821	191.707	46.438%
RF Mobile	334.236	334.236	105.146	31.459%
Power management	330.257	330.257	122.136	36.982%
Sensors and display	320.309	320.309	105.667	32.989%
Discrete and other	212.876	212.876	41.861	19.664%

2029 product totals

Product	Reference revenue	Current revenue	Current GP	Current GM
Silicon photonics including integrated optics	2,540.000	2,535.425	1,343.205	52.978%
SiGe and other RF infrastructure	458.000	458.000	211.182	46.110%
RF Mobile	380.000	380.000	120.440	31.695%
Power management	364.000	364.000	133.932	36.795%
Sensors and display	360.000	360.000	118.060	32.794%
Discrete and other	198.000	198.000	38.224	19.305%

2030 product totals

Product	Reference revenue	Current revenue	Current GP	Current GM
Silicon photonics including integrated optics	2,920.000	2,920.000	1,483.660	50.810%
SiGe and other RF infrastructure	496.000	496.000	220.563	44.468%
RF Mobile	412.000	412.000	127.916	31.048%
Power management	396.000	396.000	143.144	36.147%
Sensors and display	392.000	392.000	126.018	32.147%
Discrete and other	182.000	182.000	34.130	18.753%

Retained outer-year capacity feasibility check, reference basis: at reference Q4 2028 SiPho of USD 557.060M, a flat annual base is USD 2,228.240M. Reference FY2029 / FY2030 SiPho requires USD 311.760M / 691.760M more. At the inherited illustrative USD 6,000 per sellable 300mm wafer-equivalent, that means 51,960 / 115,293 annual additional wafers, or about 21.7% / 48.0% of a 20,000/month capacity base. That older feasibility proxy is not the USD 9,000 2027-28 expert branch ASP and is not a new multiplier. Both tests show that further capacity is already embedded in the respective reference paths; no extra Track 2 sales are added.

11. Probability and dilution sensitivities

The primary sensitivity changes only the specified expert-signal branch weights, retaining the same products, accounting basis, multiple and shares. These are diagnostic alternatives; the active case remains the selected 25% / 20% / 20% ledger.

Active ceiling sensitivity

USD per share | Q4 2027 forward window



Model figure. Active Q4 2027 price target sensitivity to the bounded expert hypotheses. USD per share; the active case applies the selected probabilities once.

Scenario	Capacity p	Pricing p	Yield p	Q4 2027 price target
Active	25%	20%	20%	215.635303
No new expert-call increment	0%	0%	0%	215.476825
Capacity branch removed	0%	20%	20%	214.178389
Competitive-pricing branch removed	25%	0%	20%	216.933740
Capacity branch alone at 100%	100%	0%	0%	221.304483
Competitive-pricing branch alone at 100%	0%	100%	0%	208.984641

Dilution sensitivity: identical operating case

Denominator change	Q4 2027 shares M	Q4 2027 price target	price target effect
-5%	110.985650	226.984529	5.26%
0%	116.827000	215.635303	0.00%
5%	122.668350	205.366955	-4.76%
10%	128.509700	196.032094	-9.09%

This isolates denominator uncertainty and does not remove known dilution. The active share schedule remains the explicit retained proxy.

12. Retained programme evidence and qualitative confidence

Programme / signal	Source and probability	Amount / unit base	Treatment in this model	Escalation / recognition
2027 SiPho commitments	Contracted delivery indication;90% confidence	1.3B inside 1.4B reference	No separate additive order layer	Accepted shipments and updated commitments.
2028 company model	Management financial model;75% execution confidence	3.6B revenue/1.63B GP/1.38B OP	Retained full-year successful-execution reference	If achieved only at exit, shift quarterly profit instead of claiming full-year delivery.
NewPhotonics volume PICs	Announced volume production;95% commercial confidence	No disclosed dollar value	Within SiPho; no duplicate sale	Recognized customer revenue and wider deployment.
6.4T product	Customer/issuer target;75% timing confidence	H12027 initial-shipment target	Inside SiPho growth, not standalone guidance	Actual shipment and qualification.
Japan Track 2	Committed plan;70% conversion confidence	20-25k/month capacity direction	Outer-year incremental wafer requirement shown above	Construction→installation→qualification→paid deliveries.
IQE supply	Signed agreement;85% availability confidence	Minimum reciprocal commitments	Supports existing products; does not eliminate all supply risk	Reliable incoming material and production yield.

The confidence percentages in this retained evidence register are qualitative analyst judgments; they are not additional arithmetic weights on the current model. Only the explicit bounded October 3 scenario ledger adds probability-weighted incremental economics. Embedded revenue is not multiplied or added a second time. Where no defensible amount or unit base exists, the opportunity remains explicitly unquantified; confidence alone cannot create a revenue number. New evidence can enlarge the TAM, improve capture or move recognition earlier, but each requires an identifiable model input.

The 75% reference execution-confidence label does not multiply the entire 2028 management case or the operating price target. The 25% / 20% / 55% mutually exclusive outcome structure and separate temporary cost stress are the explicit current economic weights.

13. Scored risk classes

Risk class	Scored deduction	Mechanism and consequence
Capacity, funding and dilution	-1	Net Japan investment 3B requires cash, grants, prepayments and qualified execution; management intent is not a financing guarantee.
Wafer pricing and competition	-1	Competing capacity or pricing concessions can change profit sharply; a higher-volume wafer mix does not ensure stable ASP.
Customer concentration	-1	Large commitments and exclusivity concentrate programme and credit exposure.
Geopolitics and supply	-1	Israel/Japan operations, trade restrictions and material supply affect timing and cost.

14. Dated catalysts, counters and escalation

1. **2026-11-15 monitoring date - Q3 results** Check 520M revenue and product mix; update actual cost absorption.
2. **2026-12-31 - SiPho run rate** Test 250M exit quarter and lag from starts to shipments.
3. **2027-06-30 - Capacity and 6.4T** Verify full effect of earlier capacity and NewPhotonics initial deliveries.
4. **2027-12-31 - Japan Track 1** Production readiness and qualifications support the 2028 full-year assumption.
5. **2028-12-31 - Track 2 and 2028 outcome** Distinguish installed tools, qualified output and full-year revenue.
6. **Every report through 2030 - Wafer price and funding** Reprice the model if ASP, project cost, shares or parent economic ownership changes.

15. Cross-book monitoring

IQE is a verified epitaxy supplier. Marvell has optical foundry relationships with both Tower and GF, so a GF agreement does not prove a Tower program loss. Integrated-laser adoption can change external-laser content per engine; the net impact on laser suppliers is not automatically positive. FAU, inspection and packaging demand grows with qualified production, subject to vendor selection. More optical volume can be positive for laser vendors while lower laser count per engine reduces content. Apply both effects; do not transfer the entire Tower growth rate to a specific upstream supplier. PCL/MSS opportunities require their own qualified process and customer evidence.

16. Source register and model limits

T1 - Q2 financial statements and Q3 outlook (2026-08-04). Financial tables distinguish GAAP from adjusted EPS and provide the diluted share anchor. Customer advances are liabilities and are not incremental revenue.

T2 - Q2 management call (2026-08-04). 2028 model: revenue USD 3.6B, GP USD 1.63B, OP USD 1.38B. Assumptions include 85% utilization, wafer prices and timely qualification. Pricing is a key sensitivity. Packaging enables SiPho rather than constituting a separately guided revenue line.

T3 - Q1 company transcript (2026-05-13). USD 1.3B SiPho customer commitments refer to wafers delivered in 2027, not wafer starts or the total opportunity.

T4 - Japan expansion announcement (2026-07-14). Track 1 targets Q4 2027 readiness and supports the 2028 model. Track 2 is expected to contribute from 2029. Approximate investment is USD 3B net of USD 1B grants.

T5 - NewPhotonics volume production filing (2026-09-17). Laser-integrated 800G/1.6T PICs are shipping in volume. The 6.4T NPC505 shipment target is H1 2027. No dollar contribution is disclosed; both sit inside SiPho.

T6 - IQE supply agreement (2026-06-15). The multi-year epiwafer agreement improves supply visibility but does not eliminate all material or qualification risks.

T7 - Conference participation notice (2026-08-12). September 10 conference participation was announced; no complete transcript is claimed here.

Q2 full-call coverage, Q1 selected passages only; no complete September conference transcript. The 2028 financial model is conditional and the full-year achievement assumption is material. Product GMs, quarterly timing and outer-year shares are LOW-confidence. Consolidated OP is not after-tax parent earnings or cash flow. No independent complete option inventory or post-restructuring minority-interest bridge was constructed.

Workbook: v3_2_master_portfolio.xlsx, model tab TSEM_Q_Model. It owns portfolio ranking, prices and return calculations. **Current artifact path:** G:/My Drive/1amInvesting/Artifacts/TSEM_TowerSemiconductor_DeepDive.pdf with a self-contained Markdown sibling. Snapshot: October 3, 2026. **Original snapshot:** versions/originals/TSEM_TowerSemiconductor_DeepDive.md. Superseded history is retained separately in VERSION_LOG, not in this current-state model.

E1 - User-supplied P Equity Research / tuolaji expert call, published October 2, 2026, 23 PDF pages. Interview date and expert identity undisclosed; AI-translated. Pages 7 / 12 concern capacity, pages 9 / 15 / 19 / 21 customer requests and allocations, page 14 yield, pages 17-20 SiGe, and pages 20-21 ST competition. This is one source chain, not multiple independent confirmations.

T8 / T9 - Official August 4, 2026 Q2 supporting slides. PDF pages 4-5 show the conditional 2028 management model; PDF page 8 shows Q2 GAAP results.

T10 - Official August 4, 2026 Q2 call transcript. PDF pages 4-5 discuss contracts and capacity conversion. The document date is August 4; later publication/crawl dates are not the call date.

T11 - July 14, 2026 Japan investor FAQ. PDF pages 2-3 distinguish Track 1 full readiness in Q4 2027, Track 2 beginning in 2029, and the scope of the 2028 targets. The combined dual-track investment is approximately USD 3B net of grants; a separate USD 500M conversion should not automatically be added on top of the same total.

T12 - May 13, 2026 SiPho contracts. USD 1.3B of 2027 contracts and USD 290M prepayments; the public release does not validate the expert's named wafer allocation split. Prepayments are financing / delivery obligations, not extra revenue.

T13 - ST March 9, 2026 PIC100 announcement. Primary evidence of 300mm SiPho competition and expansion direction; does not corroborate the expert's exact wafer prices or the conflicting annual/monthly quantities.

The issuer exhibit pixels were downloaded from the verified Tower-hosted mirror of the August 4 Q2 deck, whose cover and financial tables match the IR deck. PDF page numbering above is one-based.